

Electrical-Review Model Bakeoff

Picking the LLM for KiCraft's Layer-4 fab gate

2026-06-19 | 7 models x 3 reasoning arms (off / medium / high) x 27 designs x K=1 | 566/567 reviews | \$12.53 LLM spend | graded by Claude Code

Executive summary

We empirically compared seven mid-range LLMs as the reviewer for KiCraft's electrical-review fab gate -- the Layer-4 check that reads a finished design's netlist/BOM digest and blocks fabrication (rc7) if it finds an electrically broken-but-DRC-clean board. Each model reviewed a frozen, hand-labeled corpus (6 real blocker designs, 17 sound, 4 synthetic) at three reasoning efforts; Claude Code graders (separate context, matching-only) converted raw block flags into true recall against the labeled defects.

Bottom line

- **RECOMMENDATION:** switch the default reviewer from deepseek-v4-flash to minimax/minimax-m3 at medium reasoning effort. It is the only model that pairs perfect recall (100%) with the lowest over-block rate (14% on clean boards, 20% on warning boards) at low cost (~\$0.012/board).
- **BIGGER FINDING:** every model over-blocks. Even the best false-blocks 1-in-7 clean boards; the incumbent flash false-blocks half. A single skeptical LLM pass is a poor HARD fab gate -- the higher-value fix is to redesign the gate (cap margin/intent findings at WARNING, or require corroboration), not just swap models.
- The model ensemble cross-checked our ground-truth labels: it caught a real flaw we had missed (run_05) and exposed shared hallucinations (run_17) that no single model -- or majority vote -- would catch.

Methodology

- **Corpus:** 28 designs from a frozen self-eval snapshot, audited + datasheet-checked into 6 in-distribution blockers (DRC-clean but electrically wrong), 17 sound (8 truly clean, 9 sound-with-warnings), and 4 injected synthetics. Labels frozen + sha256'd; every model fed the IDENTICAL digest string.
- **Reviewer config** production-faithful: temperature 0, answer budget 24000 tokens (raised from 3000 -- the 2026 reasoning models emit 10-23k reasoning tokens and truncated before answering). KiCraft's cost-safety provider routing was relaxed for the review call so non-flash models route.
- **Metrics:** RECALL = of 6 blockers, fraction where the model emitted a blocker finding that semantically MATCHED the labeled defect (a defect graded warning/note = miss). FBRc = false-block rate on the 7 truly-clean boards. FBRw = block rate on the 10 warning boards. Cost/board and time/board are per-review averages.

Results -- all models x arms

Model	Arm	Rec	FBRc	FBRw	Synth	\$/bd	s/bd	p90
deepseek-v4-flash	off	83%	57%	50%	50%	\$0.0010	41	77
deepseek-v4-flash	medium	83%	71%	90%	100%	\$0.0009	38	61
deepseek-v4-flash	high	83%	43%	60%	50%	\$0.0010	34	65
deepseek-v4-pro	off	100%	57%	60%	100%	\$0.0187	96	135
deepseek-v4-pro	medium	100%	57%	60%	75%	\$0.0158	110	160
deepseek-v4-pro	high	83%	57%	80%	25%	\$0.0144	84	117
minimax-m3	off	83%	14%	40%	75%	\$0.0165	202	329
minimax-m3	medium	100%	14%	20%	75%	\$0.0125	158	388
minimax-m3	high	83%	14%	30%	50%	\$0.0138	164	289
qwen3.7-plus	off	67%	43%	40%	75%	\$0.0081	124	196
qwen3.7-plus	medium	83%	43%	70%	75%	\$0.0087	120	191
qwen3.7-plus	high	83%	29%	50%	50%	\$0.0082	116	185

glm-5.2	off	100%	43%	60%	75%	\$0.0321	165	272
glm-5.2	medium	100%	29%	40%	75%	\$0.0367	156	366
glm-5.2	high	100%	14%	40%	75%	\$0.0361	190	422
mistral-medium-3-5	off	83%	100%	100%	100%	\$0.0107	11	22
mistral-medium-3-5	medium	83%	100%	100%	100%	\$0.0774	82	161
mistral-medium-3-5	high	83%	100%	100%	100%	\$0.0704	64	100
claude-haiku-4.5	off	33%	100%	100%	100%	\$0.0095	14	18
claude-haiku-4.5	medium	67%	43%	70%	100%	\$0.0308	46	66
claude-haiku-4.5	high	67%	57%	70%	75%	\$0.0407	62	83

Rec = recall on 6 blockers; FBRc/FBRw = false-block on clean / warning boards (lower is better); \$/bd and s/bd = average cost and wall-time per review. Best arm per model is shaded.

Per-model breakdown

deepseek/deepseek-v4-flash (incumbent)

best arm: high recall 83% FBRc 43% FBRw 60% \$0.0010/board 34s/board (p90 65s)

The incumbent: by far the cheapest (\$0.001) and fastest (~35s). Catches 5 of 6 blockers and never fails to return JSON. But it over-blocks 43-71% of clean boards, and -- unusually -- reasoning makes it WORSE (medium-effort FBRc 71% vs off 57%): extra deliberation makes the small model more trigger-happy. Misses the subtle R-2R topology defect and only flags half the synthetic floor on its off/high arms.

deepseek/deepseek-v4-pro

best arm: off recall 100% FBRc 57% FBRw 60% \$0.0187/board 96s/board (p90 135s)

Full recall on its off/medium arms, but over-blocks 57% of clean boards with no improvement from reasoning. Reasons heavily (~20k tokens), so it is slow (~100s) and 10-20x flash's cost. Good detector, poor precision -- not worth the price over minimax.

minimax/minimax-m3

best arm: medium recall 100% FBRc 14% FBRw 20% \$0.0125/board 158s/board (p90 388s)

The winner. At medium effort it is the only model with 100% recall AND the lowest over-block (14% clean / 20% warning -- best on both), at a modest \$0.012/board. Reasoning helps it (off->medium lifts recall 83->100%). Downsides: slow (~2.5 min average, p90 6.5 min) and it can over-reason on complex digests -- it stalled once (run_18, off arm, ~15 min, no answer). At medium it was reliable; the gate is fail-soft so a stall degrades to 'skipped', not a hung build.

qwen/qwen3.7-plus

best arm: high recall 83% FBRc 29% FBRw 50% \$0.0082/board 116s/board (p90 185s)

The cheap-and-decent option (\$0.008). High-effort gives 83% recall / 29% FBRc -- weaker recall than minimax/glm but the best precision among the truly low-cost models. A reasonable budget pick if minimax's latency is unacceptable and flash over-blocks too much.

z-ai/glm-5.2

best arm: high recall 100% FBRc 14% FBRw 40% \$0.0361/board 190s/board (p90 422s)

The strongest alternative to minimax: 100% recall at every arm, and high effort drops FBRc to 14% (reasoning clearly helps it: 43->14% off->high). More consistent than minimax (no stalls). But 3x the cost (\$0.036) and 2x the warning-board over-block (40% vs 20%), and the slowest p90 (7 min). The recommended fallback if minimax's reliability/latency disqualifies it.

mistralai/mistral-medium-3-5

best arm: off recall 83% FBRc 100% FBRw 100% \$0.0107/board 11s/board (p90 22s)

Unusable as a gate: it blocks 100% of clean boards at every effort -- it would reject every good design. (It also rejects the production reasoning={max_tokens} format with a 400, only accepting effort.) High recall is meaningless without precision.

anthropic/claude-haiku-4.5 (ref)

best arm: medium recall 67% FBRc 43% FBRw 70% \$0.0308/board 46s/board (p90 66s)

Reference only (not production-eligible). Poor: off-arm catches just 33% of blockers while blocking 100% of clean boards; reasoning helps (medium 67% recall / 43% FBRc) but it never becomes competitive, at \$0.01-0.04/board.

Notable successes & failures

- Real defect the ensemble caught that we MISLABELED: run_05 (USB-PD trigger) -- 20/21 cells flagged that the CH224K's open-drain power-good pin drives the indicator LED backwards (it can never light). A genuine flaw we missed (warning-severity; we reclassified the label).
- Shared hallucination that corroboration CANNOT fix: run_17 (AL8860 LED driver) -- 17/21 cells 'found' a 0.2V sense reference setting 2A on a 1.5A part. The part is 0.1V -> 1A (sound, user-confirmed). A majority

vote would have wrongly blocked it.

- Scattered hallucinations (filtered by corroboration): run_04 crossover (4/21, botched the filter math -- 10uF IS correct for 2kHz), run_01 RC filter (5/21, a valid adjustable topology).
- Recurring single-defect hallucination: run_22 (motor board) -- many models 'found' an incomplete feedback divider on the AP63203, which is a FIXED 3.3V part. Graders correctly routed these to 'unmatched', so they did not inflate recall.
- Hardest real blocker: run_02 (R-2R DAC) -- only 12/21 matched the labeled topology defect; several models instead flagged a different real issue (missing ladder termination / op-amp feedback), showing recall is sensitive to which of multiple true faults a model fixates on.
- Easiest real blockers: the structural/value faults -- A4988 sense resistors (run_27, 19/21), TPS5430 wrong feedback voltage (run_15, 19/21), no-firmware-path (run_22, 19/21), relay missing input connector (run_19, 20/21). Models reliably catch concrete, citable defects.

Conclusion & recommendation

Set KICRAFT_REVIEW_MODEL = minimax/minimax-m3 with reasoning effort = medium. Across the corpus it is the single best reviewer: 100% recall on real blockers, the lowest false-block rate of any model (14% clean / 20% warning), at ~\$0.012 per board. It strictly dominates the incumbent flash, which catches fewer blockers (83%) and false-blocks far more (43-71% of clean boards). z-ai/glm-5.2 at high effort is the recommended fallback (equal recall and clean-FBR, more reliable, but 3x the cost, 2x the warning over-block, and slower).

The cost of the switch is latency: minimax averages ~2.5 minutes per review and can reach ~6.5 minutes, versus ~35 seconds for flash. Because the gate runs once per build, is fail-soft (any error/timeout degrades to 'skipped', never a hung build), and is opt-out via KICRAFT_ELECTRICAL_REVIEW=0, this is an acceptable trade for materially better defect-catching and far fewer wrongly-rejected good designs. The answer budget (review_max_tokens=24000) and review-only routing relaxation are already in place to support it.

However -- the dominant result is that EVERY model over-blocks. The best still false-blocks 1-in-7 clean boards and 1-in-5 warning boards; a single skeptical LLM pass is fundamentally too aggressive for a HARD fab gate. The highest-value follow-up is therefore to redesign the gate, not just the model: cap margin / intent / sizing findings at WARNING and hard-block only on the unambiguous classes (reversed power, no programming path, dead short, grossly out-of-spec rail), and/or require ≥ 2 -model corroboration before a hard block. Switching to minimax is the immediate win; redesigning the gate is the structural fix.

Caveats

- Small N: 6 blockers + 7 clean -> recall and FBRc resolve to about +/-1 case (~15%); rankings within ~15% are ties (minimax-medium and glm-high are statistically tied on recall+clean-FBR).
- K=1: one sample per cell, so temperature-0 consistency was not measured. A follow-up $K \geq 3$ run on the top 2-3 models would quantify flakiness.
- Single corpus: one self-eval snapshot, one design model's digest style. Confirm on a second snapshot before locking the model long-term.
- Our hand-labels were imperfect (the ensemble caught run_05); treat the absolute FBR numbers as indicative, the relative model ranking as robust.